

Tanvir Ahmed

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RESEARCH INTERESTS

Cyber Security • Reinforcement Learning • Quantum Machine Learning • Computer Vision

RESEARCH SUMMARY

Computer science graduate and independent researcher in machine learning, working mainly on security for networked systems. My accepted work builds a real-time browser threat detection agent that runs a vision-language model and an LLM locally, so pages are screened on consumer hardware without sending anything to a server. A second security line treats intrusion-detection feature selection as a combinatorial optimization problem, solved with QAOA on IBM Quantum hardware and with simulated annealing, then benchmarked against classical selectors on industrial IoT traffic (X-IIoTID, NF-ToN-IoT, NF-UNSW-NB15). Related work detects and repairs the damage feature selection does to rare attack classes. Alongside this I work on quantum machine learning and on deep learning for medical images. I am now moving towards reinforcement learning for adaptive defence, where an agent has to choose a response under uncertainty rather than score one flow at a time. Code is public for every project.

EDUCATION

Independent University, Bangladesh

January 2015 — December 2019

Department of Computer Science and Engineering

Bachelor of Science in Computer Science, Minor in Environmental Management

CGPA: 2.92 out of 4.00 (Final-year GPA: 3.07)

PUBLICATIONS

Journal paper

Under Review — Major Revision Submitted

- Abhijit Kumar Ghosh, Ar Rafiul Islam Rafim, Fahmid Al Farid, **Tanvir Ahmed**, Enam Ahmed Taufik, Jia Uddin, and Hezerul Abdul Karim. “Gated Dual-Scale Feature Fusion for Colorectal Cancer Grading in Centralized and Federated Pathology.” *Informatics in Medicine Unlocked*, 2026. Major revision submitted. Code: [GitHub].

Conference paper

Accepted

- Dhiman Kumar Sarker, Fyaz Nafin Rahman, Md Rezaul Karim Khan, Sushmoy Paul Rohit, **Tanvir Ahmed**, and Md Mizanur Rahman. “Real-Time Browser Threat Detection Security Agent via Local VLM-LLM.” Accepted at the *5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet-of-Things (RAAICON 2026)*, 2026.

Submitted

- Tanvir Ahmed**, Jannatul Ferdous Shorna, Md Mahbubur Akash, Md Khorshed Alam, and Md Tarek Hasan. “DARE-FS: A Selector-Agnostic Operator for Detecting and Repairing Class-Specific Feature-Selection Harm.” Manuscript submitted to the *38th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2026)*, 2026. Paper ID: 778.
- Swagotam Malakar, **Tanvir Ahmed**, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Beyond Confidence: A Leakage-Audited Selective-Prediction Study of Acid-Fast Bacilli Detection in Ziehl-Neelsen Microscopy.” Manuscript submitted to the *International Conference on Digital Image Computing: Techniques and Applications (DICTA 2026)*, 2026. Paper ID: 79.
- Md Faisal Sheikh, **Tanvir Ahmed**, Enam Ahmed Taufik, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Towards Trustworthy Tuberculosis Detection: Subgroup Calibration of Object Detectors in Sputum-Smear Microscopy.” Manuscript submitted to the *International Conference on Digital Image Computing: Techniques and Applications (DICTA 2026)*, 2026. Paper ID: 80.

PROJECTS

QCIDS: Quantum-Classical Feature Selection for Network Intrusion Detection [GitHub]

- Cast intrusion-detection feature selection as a QUBO solved by QAOA ($p = 1, 2, 3$) on simulators and IBM 156-qubit hardware, simulated annealing, and classical baselines under matched leakage-free cross-validation; at $k = 15$ on NF-ToN-IoT, QAOA $p = 2$ held macro-F1 at 0.763 where greedy mutual information degraded from 0.760 to 0.718.

CovNCG: Training-Free Dense Quantum Feature Maps [GitHub]

- Designed a training-free quantum kernel from non-commuting multi-feature generator blocks (three features per two qubits) with a closed-form kernel-variance result scaling as c^4 , turning avoidance of exponential concentration into a bandwidth tuning rule; evaluated with a projected kernel and SVM on five medical datasets in Qiskit with IBM QPU validation.

Q-PlasmaNet: Hybrid Classical–Quantum Classifier for Plasma Cell Recognition [GitHub]

- Paired a frozen CNN backbone with a 4–8 qubit data re-uploading variational quantum head (PennyLane) at matched parameter budget, reaching F1 0.927 with 178 trainable parameters — statistically tied with a parameter-matched MLP (0.932) and below a 2.2M-parameter MobileNetV2 (0.949), establishing parity rather than quantum advantage.

PCMMD: Federated Learning and Knowledge Distillation for Plasma Cell Detection [GitHub]

- Benchmarked centralized, few-shot, knowledge-distillation (EfficientNet-B0 teacher to MobileNetV3 student), and federated (FedAvg/FedProx, one client per patient) regimes for plasma-cell classification of bone-marrow crops in multiple myeloma, under patient-disjoint folds, validation-only model selection, and image-hash de-duplication to eliminate leakage.

Direct-WAM: Self-Supervised Denoising for Two-Photon Calcium Imaging [GitHub]

- Proved a bias-floor theory of self-supervised objectives (per-objective error floor, ensemble bias-crossing, distillation transfer bound) and produced a corrected AI4Life CIDC25 benchmark in which a faithfully retrained DeepCAD-RT reaches +23.65 dB stSNR, while the F_0 -free Direct-WAM 3D U-Net distilled from a 13-way ensemble reaches +18.15 dB.

RESEARCH EXPERIENCE**Independent Researcher***Independent and collaborative research*

Boca Raton, Florida, USA

January 2025 — Present

- Author and co-author five manuscripts with international collaborators, one accepted and four under review, across medical imaging, quantum machine learning and feature selection.
- Run the full experimental pipeline on each project: auditing the data for leakage, building and training the models, benchmarking against classical and quantum baselines, and reporting results over multiple seeds with confidence intervals.

PROFESSIONAL EXPERIENCE**Career Break — Relocation to the United States***F-2 dependent status*

Boca Raton, Florida, USA

January 2023 — December 2024

- Relocated from Bangladesh to the United States as an F-2 dependent, a status that does not authorize employment; used the period for self-directed study in machine learning, which led into the independent research work above.

Upwork & Fiverr*Freelance Web Developer and E-commerce Specialist*

Remote — Dhaka, Bangladesh

December 2014 — November 2022

- Delivered e-commerce storefronts for international clients, both as customized Shopify builds (themes, product catalogues, checkout flows) and as custom-developed websites written from scratch to client specification.
- Ran day-to-day store operations across order and inventory management, customer inquiries, and issue resolution, sustaining repeat client relationships over an eight-year freelance practice.

IDOL Focus — A Professional Training Institute*Web Developer Intern*

Dhaka, Bangladesh

September 2019 — December 2019

Supervisors: Md. Asif Bin Khaled, Lecturer, Department of Computer Science and Engineering, IUB (academic); Md. Al Mamun, IDOL Focus (industry)

- Built the Admin and Agent modules of an online job portal (*idolfocusjobs.com*) serving four user roles: role-based authentication, jobseeker and employer account management, CV database administration, job-posting approval, and payment handling.
- Implemented in PHP, JavaScript, Ajax, jQuery, and Bootstrap over MySQL; documented in an undergraduate internship report submitted toward the BSc degree.

TECHNICAL SKILLS

Security:	Network intrusion detection, IIoT traffic analysis, threat detection agents, data-leakage auditing
Programming:	Python, SQL, PHP, JavaScript
ML Frameworks:	PyTorch, Scikit-learn, LightGBM, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib
Quantum ML:	PennyLane, Qiskit, QAOA/QUBO formulation, variational circuits, quantum kernels, IBM Quantum hardware
Core Areas:	Machine Learning, Computer Vision, Deep Learning, Federated Learning, Knowledge Distillation
Databases:	MySQL, phpMyAdmin
Web / Software:	HTML, CSS, JavaScript, jQuery, Ajax, Bootstrap, Shopify, Apache Web Server
Tools:	L ^A T _E X, Git, Jupyter Notebook, Google Colab
Languages:	Bangla (native), English (fluent)